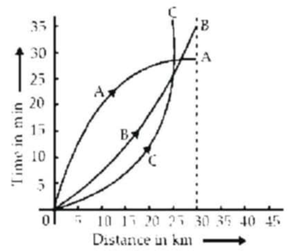
Hence,

Option A, B and C is correct and option D "Both trains travel the same distance in time to units" is not correct.

53. Consider the following distance-time graphs. The graph shows three athletes A,B and C running side by side for a 30 km race.



With reference to the above graph, consider the following statements :

1. The race was won by A .
2. B was ahead of A up to 25 km mark .
3. C ran very slowly from the beginning.

Which of the statements given above is /are correct?

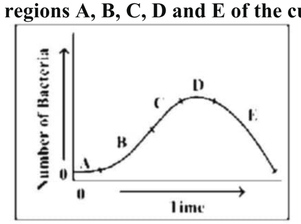
(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Ans. (b) With reference to the given graph, 1 and 2 statements are correct.

Directions for the following 3 (three) items

Read the passage given below, study the graph that follows and answer the three items given below the figure.

During a party, a person was exposed to contaminated water. A few days later, he developed fever and loose motions. He suffered from some days before going to a doctor for treatment. On starting the treatment, he soon became better and recovered completely a few days later. The following graph shows different phases of the person's disease condition as regions A, B, C, D and E of the curve.



54. Which region/regions of the curve correspond/ corresponds to incubation phase of the infection?

(a) A only
(b) B only
(c) B and C
(d) No part of the curve indicates the incubation phase

Ans. (a) Region 'A' of the curve corresponds to incubation phase of the infection.

55. Which region of the curve indicates that the person began showing the symptoms of infection?

(a) A (b) B
(c) C (d) D

Ans. (b) Region 'B' of the curve indicates that the person began showing the symptoms of infection.

56. Which region of the curve indicates that the treatments yielded effective relief ?

(a) C
(b) D
(c) E
(d) The curve does indicate the treatment

Ans. (c) Region 'E' of the curve indicates that the treatment yielded effective relief.

57. There are four routes to travel from city A to city B and six routes from city B to city C. How many routes are possible to travel from the city A to city C ?

(a) 24 (b) 12
(c) 10 (d) 8

Ans. (a) Possible routes to travel from the city A to city C = 4 × 6

$\Rightarrow$ 24

58. A contract on construction job specifies a penalty for delay in completion of the work beyond a certain date is as follows : Rs. 200 for the first day etc. Rs. 250 for the second day Rs. 300 for the third day etc. the penalty for each succeeding day being Rs.50 more than that of the preceding day. How much penalty should the contractor pay if he delays the work by 10 days ?

(a) ₹ 4950 (b) ₹ 4250
(c) ₹ 3600 (d) ₹ 650

Ans. (b) According to question, we find the arithmetic progression,
200, 250, 300 ..... (till 10 days)

$\because$ a = 200
d = 50
n = 10

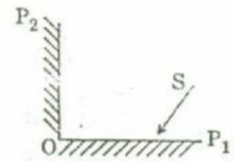
$$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$$

$$= \frac{10}{2}[2 \times 200 + (10-1)\,50]$$

$$= 5\,[400 + 450]$$

$$= 5\ (850) = 4250\ ₹$$

59. Consider the figure given below and answer the item that follows :

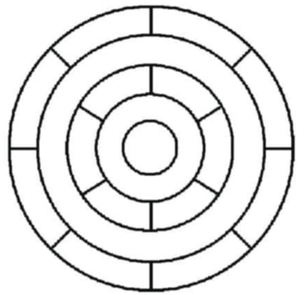


In the figure shown, OP_1 and OP_2 are two plane mirrors kept perpendicular to each other. S is the direction of a beam of light falling on the mirror OP_1 the direction of the reflected beam of ligh from the mirror OP_2 wills

(a) Perpendicular to the direction S.
(b) At 45^0 to the direction S.
(c) Opposite and parallel to the direction S.
(d) At 60^0 to the directions S.

Ans. (c) The direction of the reflected beam of light from the mirror OP_2 wills opposite and parallel to direction S.

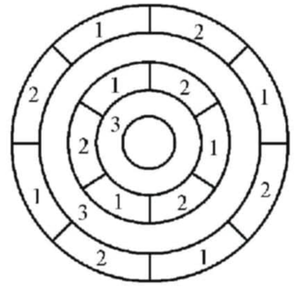
60. Consider the following figure and answer the item that follows :



What is the minimum number of different colours required to paint the figure given above such that no two adjacent regions have the same colour?

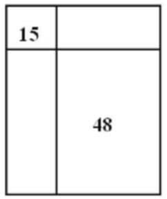
(a) 3 (b) 4
(c) 5 (d) 6

Ans. (a)



The minimum number of different colours required to paint the figure = 3

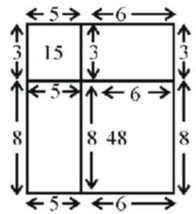
61. Consider the following figure and answer the item that follows :



A square is divided into four rectangles as shown above. The lengths of the side of rectangles are natural numbers. The areas of two rectangles are indicated in the figure.

(a) 10
(b) 11
(c) 15
(d) Cannot be determined as the given data are insufficient.

Ans. (b)



The length of the square = 11

62. Three persons start walking together and their steps measure 40 cm, 42 cm and 45 cm respectively. What is the minimum distance ech should walk so that each can cover the same distance in complete steps?